Unsupervised Learning

100 Questions: K-means, Hierarchical, DBSCAN, Evaluation Metrics

GARP RAI Exam Preparation

Q1: Unsupervised Learning Definition - Tutorial

QUESTION: What is unsupervised learning?

OPTIONS:

1. Learning with labeled data where the outcome is known
2. Learning with unlabeled data to find hidden patterns and structures
3. Learning that only uses numerical features
4. Learning that requires human supervision

ANSWER: B

DETAILED EXPLANATION:

- Unsupervised learning navigates datasets with only features/characteristics
- No predefined labels or outcomes
- Goal: uncover hidden patterns, structures, or insights
- Primary application: clustering/segmentation

WHY OTHER OPTIONS ARE WRONG:

- A: This describes SUPERVISED learning
- C: Unsupervised can use any type of features
- D: Unsupervised learning does NOT require human supervision

EXAM TRAP: Unsupervised learning = unlabeled data to find patterns!

Q2: Clustering Definition - Tutorial

QUESTION: What is the fundamental goal of clustering analysis?

OPTIONS:

1. To predict future outcomes
2. To group data points so similar items are together and different items are apart
3. To reduce the number of features
4. To classify data into predefined categories

ANSWER: B

DETAILED EXPLANATION:

- Arrange data points into groups (clusters)
- Within-group similarity is high
- Between-group similarity is low
- Also called segmentation

WHY OTHER OPTIONS ARE WRONG:

- A: This is prediction/regression (supervised)
- C: This is dimensionality reduction
- D: This is classification (supervised)

EXAM TRAP: Clustering = group similar items together!

Q3: Clustering Applications - Tutorial

QUESTION: Which is an application of clustering in risk management?

OPTIONS:

1. Customer risk profiling
2. Predictive maintenance
3. Cybersecurity threat identification
4. All of the above

ANSWER: D

DETAILED EXPLANATION:

- **Customer risk profiling:** Identify customer segments for tailored strategies
- **Operational risk:** Detect anomalies in machinery sensor data
- **Cybersecurity:** Identify unusual network traffic patterns
- **Regulatory compliance:** Group similar documents for review

WHY OTHER OPTIONS ARE WRONG:

- A: Correct but incomplete
- B: Correct but incomplete
- C: Correct but incomplete

EXAM TRAP: Clustering has many risk management applications!

Q4: Clustering as Preliminary Step - Tutorial

QUESTION: Why might clustering be used before supervised learning?

OPTIONS:

1. To increase the number of features
2. To understand the inherent structure of features
3. To remove all labels
4. To make the data smaller

ANSWER: B

DETAILED EXPLANATION:

- Temporarily ignore labels to understand data structure
- Gain insights into feature characteristics
- Can improve supervised model performance
- Exploratory data analysis

WHY OTHER OPTIONS ARE WRONG:

- A: Clustering doesn't increase features
- C: Labels are only temporarily ignored
- D: Clustering doesn't reduce data size

EXAM TRAP: Clustering helps understand feature structure before supervised learning!

Q5: Data Scaling Importance - Tutorial

QUESTION: Why is data scaling critical before clustering?

OPTIONS:

1. To make data categorical
2. Because all clustering methods rely on distance/similarity measures
3. To remove all outliers
4. To increase data size

ANSWER: B

DETAILED EXPLANATION:

- Features on different scales skew distance calculations
- Large-scale features disproportionately influence results
- Example: Age (years) vs Wealth (hundreds of thousands)
- Standardization or normalization ensures fair contribution

WHY OTHER OPTIONS ARE WRONG:

- A: Scaling doesn't make data categorical
- C: Scaling doesn't remove outliers
- D: Scaling doesn't increase data size

EXAM TRAP: Scaling is essential because clustering uses distance measures!

Q6: Partitional Clustering - Tutorial

QUESTION: What is partitional clustering?

OPTIONS:

1. Clustering that creates a tree-like structure
2. Clustering that divides data into a predetermined number of non-overlapping groups
3. Clustering based on density of data points
4. Clustering that starts with each point as its own cluster

ANSWER: B

DETAILED EXPLANATION:

- Each data point belongs to a single cluster
- Non-overlapping groups
- Example: K-means clustering
- Based on proximity to centroid

WHY OTHER OPTIONS ARE WRONG:

- A: This is hierarchical clustering
- C: This is density-based clustering
- D: This is agglomerative hierarchical clustering

EXAM TRAP: Partitional clustering = non-overlapping groups with predetermined K!

Q7: Hierarchical Clustering - Tutorial

QUESTION: What does hierarchical clustering construct?

OPTIONS:

1. A flat structure of clusters
2. A tree-like structure (dendrogram) of clusters
3. A set of centroids
4. A density map

ANSWER: B

DETAILED EXPLANATION:

- Builds a hierarchy of clusters
- Visualized as a dendrogram
- Two approaches: divisive and agglomerative
- Does not require pre-specifying K

WHY OTHER OPTIONS ARE WRONG:

- A: This is partitional clustering
- C: This is K-means centroids
- D: This is density-based clustering

EXAM TRAP: Hierarchical clustering creates tree-like dendrograms!

Q8: Divisive Clustering - Tutorial

QUESTION: What is divisive clustering?

OPTIONS:

1. Bottom-up approach starting with each point as its own cluster
2. Top-down approach starting with all points in one cluster
3. Clustering based on density
4. Clustering with predefined number of clusters

ANSWER: B

DETAILED EXPLANATION:

- Top-down approach
- All data points in a single cluster initially
- Iteratively splits into smaller, more distinct clusters
- Continues until stopping criterion is met

WHY OTHER OPTIONS ARE WRONG:

- A: This is agglomerative clustering
- C: This is density-based clustering
- D: This is partitional clustering

EXAM TRAP: Divisive = top-down (split from one cluster)!

Q9: Agglomerative Clustering - Tutorial

QUESTION: What is agglomerative clustering?

OPTIONS:

1. Top-down approach starting with all points in one cluster
2. Bottom-up approach starting with each point as its own cluster
3. Clustering based on density of points
4. Clustering with predefined centroids

ANSWER: B

DETAILED EXPLANATION:

- Bottom-up approach
- Each data point starts as its own cluster
- Merges closest clusters iteratively
- Continues until one cluster remains

WHY OTHER OPTIONS ARE WRONG:

- A: This is divisive clustering
- C: This is density-based clustering
- D: This is partitional clustering

EXAM TRAP: Agglomerative = bottom-up (merge from individual points)!

Q10: Density-Based Clustering - Tutorial

QUESTION: How does density-based clustering define clusters?

OPTIONS:

1. By distance to centroids
2. By dense regions of data points separated by low-density regions
3. By hierarchical merging
4. By predefined number of groups

ANSWER: B

DETAILED EXPLANATION:

- Defines clusters as dense regions
- Separated by regions of lower point density
- Can find non-spherical clusters
- More robust to outliers
- Example: DBSCAN

WHY OTHER OPTIONS ARE WRONG:

- A: This is K-means/partitional
- C: This is hierarchical
- D: This is partitional

EXAM TRAP: Density-based clustering = dense regions separated by low density!

Q11: K-means Definition - Tutorial

QUESTION: What is the K-means algorithm?

OPTIONS:

1. A hierarchical clustering method
2. A partitional clustering method that groups data into K clusters based on centroids
3. A density-based clustering method
4. A classification algorithm

ANSWER: B

DETAILED EXPLANATION:

- Partitions N observations into K clusters
- Groups based on proximity to centroids
- Iterative process
- Requires pre-specifying K

WHY OTHER OPTIONS ARE WRONG:

- A: K-means is partitional, not hierarchical
- C: K-means is centroid-based, not density-based
- D: K-means is unsupervised clustering, not classification

EXAM TRAP: K-means = partitional clustering based on centroids!

Q12: K-means Requirement - Tutorial

QUESTION: What must be specified before running K-means?

OPTIONS:

1. The number of features
2. The number of clusters K
3. The distance metric only
4. The number of iterations

ANSWER: B

DETAILED EXPLANATION:

- K (number of clusters) must be specified
- Analyst determines this beforehand
- Often experiment with different K values
- Use evaluation techniques to select optimal K

WHY OTHER OPTIONS ARE WRONG:

- A: Features are given, not specified
- C: Distance metric is chosen, but K must be specified
- D: Iterations are determined by convergence

EXAM TRAP: K-means requires pre-specifying K (number of clusters)!

Q13: K-means Algorithm Steps - Tutorial

QUESTION: What are the iterative steps of K-means?

OPTIONS:

1. Assign points to nearest centroid, update centroids, repeat
2. Calculate WCSS, find elbow, choose K
3. Create dendrogram, merge clusters
4. Identify core points, border points, noise

ANSWER: A

DETAILED EXPLANATION:

- **Step 1:** Randomly select K initial centroids
- **Step 2:** Assign each data point to nearest centroid
- **Step 3:** Update centroids to mean of assigned points
- **Step 4:** Repeat until convergence

WHY OTHER OPTIONS ARE WRONG:

- B: This is for choosing K, not the algorithm
- C: This is hierarchical clustering
- D: This is DBSCAN

EXAM TRAP: K-means = assign $\rightarrow$ update $\rightarrow$ repeat!

Q14: K-means Convergence - Tutorial

QUESTION: When does the K-means algorithm stop?

OPTIONS:

1. After a fixed number of iterations
2. When centroids no longer change or points stop switching clusters
3. When WCSS reaches zero
4. When all points are in one cluster

ANSWER: B

DETAILED EXPLANATION:

- Algorithm has converged
- Centroids stabilize
- Cluster assignments no longer change
- Called Lloyd's algorithm

WHY OTHER OPTIONS ARE WRONG:

- A: Converges based on stability, not iterations
- C: WCSS reaches zero only when K=N
- D: This would be one cluster (K=1)

EXAM TRAP: K-means converges when centroids stabilize and assignments stop changing!

Q15: Euclidean Distance Formula - Tutorial

QUESTION: What is the Euclidean distance between two points?

OPTIONS:

1. $d_E = |x_1 - x_2| + |y_1 - y_2|$
2. $d_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
3. $d_E = (x_1 - x_2)^2 + (y_1 - y_2)^2$
4. $d_E = \max(|x_1 - x_2|, |y_1 - y_2|)$

ANSWER: B

DETAILED EXPLANATION:

- Straight-line ("as the crow flies") distance
- Also called L2-norm
- Sum of squared differences, then square root
- Most common distance measure in K-means

WHY OTHER OPTIONS ARE WRONG:

- A: This is Manhattan distance (L1-norm)
- C: This is squared distance without square root
- D: This is Chebyshev distance

EXAM TRAP: Euclidean distance = $\sqrt{\sum(x_i - y_i)^2}$!

QUESTION: What is the Manhattan distance between two points?

OPTIONS:

1. $d_M = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
2. $d_M = |x_1 - x_2| + |y_1 - y_2|$
3. $d_M = (x_1 - x_2)^2 + (y_1 - y_2)^2$
4. $d_M = \max(|x_1 - x_2|, |y_1 - y_2|)$

ANSWER: B

DETAILED EXPLANATION:

- Also called L1-norm
- Sum of absolute differences
- Like driving in a city with rectangular grid
- Alternative to Euclidean distance

WHY OTHER OPTIONS ARE WRONG:

- A: This is Euclidean distance
- C: This is squared distance
- D: This is Chebyshev distance

EXAM TRAP: Manhattan distance = sum of absolute differences!

Q17: Euclidean vs Manhattan - Tutorial

QUESTION: Which distance measure is also known as L1-norm?

OPTIONS:

1. Euclidean distance
2. Manhattan distance
3. Chebyshev distance
4. Cosine distance

ANSWER: B

DETAILED EXPLANATION:

- Manhattan distance = L1-norm
- Euclidean distance = L2-norm
- L1-norm: sum of absolute differences
- L2-norm: square root of sum of squared differences

WHY OTHER OPTIONS ARE WRONG:

- A: Euclidean is L2-norm
- C: Chebyshev is Linfinity-norm
- D: Cosine is based on angles

EXAM TRAP: Manhattan = L1-norm; Euclidean = L2-norm!

Q18: K-means Example - Assignment Step - Tutorial

QUESTION: In K-means example, what determines which cluster a point belongs to?

OPTIONS:

1. Random assignment
2. The closer centroid based on distance
3. The centroid with larger coordinates
4. The first centroid encountered

ANSWER: B

DETAILED EXPLANATION:

- Each point assigned to nearest centroid
- Distance measured using Euclidean or Manhattan
- Point assigned to cluster with minimum distance
- Creates Voronoi partitioning of space

WHY OTHER OPTIONS ARE WRONG:

- A: Assignment is deterministic based on distance
- C: Distance to centroid determines assignment
- D: Assignment is based on distance, not order

EXAM TRAP: Assignment = nearest centroid based on distance!

Q19: K-means Example - Update Step - Tutorial

QUESTION: How is the centroid updated in K-means?

OPTIONS:

1. Randomly repositioned
2. Set to the mean/average of all points in its cluster
3. Set to the median of all points in its cluster
4. Moved to the farthest point in its cluster

ANSWER: B

DETAILED EXPLANATION:

- Centroid becomes mean of assigned points
- Average of all features for points in cluster
- Minimizes WCSS for that cluster
- New centroid position reflects cluster center

WHY OTHER OPTIONS ARE WRONG:

- A: Update is deterministic, not random
- C: K-means uses mean, not median
- D: Centroid moves to center, not farthest point

EXAM TRAP: Centroid update = mean of points in cluster!

Q20: K-means Example - Iteration - Tutorial

QUESTION: What happens during each iteration of K-means?

OPTIONS:

1. Only centroids are updated
2. Points are assigned and centroids are updated
3. Only points are assigned
4. Random points are selected

ANSWER: B

DETAILED EXPLANATION:

- Each iteration has two steps
- **Assignment step:** Points assigned to nearest centroid
- **Update step:** Centroids recalculated
- Repeated until convergence

WHY OTHER OPTIONS ARE WRONG:

- A: Both assignment and update happen
- C: Both assignment and update happen
- D: Selection happens only at initialization

EXAM TRAP: Each iteration = assignment + update!

Q21: WCSS Definition - Tutorial

QUESTION: What does WCSS (Within-Cluster Sum of Squares) measure?

OPTIONS:

1. Distance between clusters
2. Compactness of clusters (how close points are to their centroids)
3. Number of data points
4. Overall variance of data

ANSWER: B

DETAILED EXPLANATION:

- Sum of squared distances from each point to its centroid
- Lower WCSS = more compact clusters
- Also called inertia
- Key metric for evaluating K-means performance

WHY OTHER OPTIONS ARE WRONG:

- A: This is BCSS (Between-Cluster Sum of Squares)
- C: This is count of data points
- D: This is total variance

EXAM TRAP: WCSS = compactness of clusters (within-cluster distance)!

Q22: WCSS Formula - Tutorial

QUESTION: What is the formula for WCSS of a single cluster?

OPTIONS:

1. $WCSS_k = \sum_{j=1}^{n_k} d_j$
2. $WCSS_k = \sum_{j=1}^{n_k} d_j^2$
3. $WCSS_k = \frac{1}{n_k} \sum_{j=1}^{n_k} d_j$
4. $WCSS_k = \sum_{j=1}^{n_k} |d_j|$

ANSWER: B

DETAILED EXPLANATION:

- d_j = distance from point j to centroid
- Sum of squared distances
- n_k = number of points in cluster k
- Squaring penalizes larger distances

WHY OTHER OPTIONS ARE WRONG:

- A: Uses squared distances, not just distances
- C: This is average distance, not sum
- D: Uses squared, not absolute

EXAM TRAP: WCSS = sum of squared distances to centroid!

Q23: Total WCSS Formula - Tutorial

QUESTION: How is total WCSS calculated for all clusters?

OPTIONS:

1. $WCSS = \sum_{k=1}^{K} WCSS_k$
2. $WCSS = \frac{1}{K} \sum_{k=1}^{K} WCSS_k$
3. $WCSS = \max(WCSS_k)$
4. $WCSS = \min(WCSS_k)$

ANSWER: A

DETAILED EXPLANATION:

- Sum of WCSS across all K clusters
- Measures overall compactness
- Total within-cluster variation
- Used for scree plot and evaluation

WHY OTHER OPTIONS ARE WRONG:

- B: This is average WCSS
- C: This is maximum WCSS
- D: This is minimum WCSS

EXAM TRAP: Total WCSS = sum of WCSS over all clusters!

Q24: BCSS Definition - Tutorial

QUESTION: What does BCSS (Between-Cluster Sum of Squares) measure?

OPTIONS:

1. Compactness of clusters
2. Separation between clusters
3. Number of data points
4. Within-cluster variation

ANSWER: B

DETAILED EXPLANATION:

- Sum of squared distances of cluster centroids to overall centroid
- Weighted by number of points in each cluster
- Higher BCSS = better separated clusters
- Measures inter-cluster dissimilarity

WHY OTHER OPTIONS ARE WRONG:

- A: This is WCSS (within-cluster)
- C: This is count of data points
- D: This is within-cluster variation

EXAM TRAP: BCSS = separation between clusters!

Q25: WCSS and BCSS Relationship - Tutorial

QUESTION: What is the relationship between WCSS and BCSS?

OPTIONS:

1. Minimizing WCSS implies maximizing BCSS
2. They are independent
3. Maximizing WCSS implies maximizing BCSS
4. There is no relationship

ANSWER: A

DETAILED EXPLANATION:

- More compact clusters (low WCSS) tend to be more separated (high BCSS)
- Total sum of squares = WCSS + BCSS (for fixed data)
- Minimizing WCSS implicitly maximizes BCSS
- Both are important for good clustering

WHY OTHER OPTIONS ARE WRONG:

- B: They are related through total variance
- C: Maximizing WCSS would mean poor clusters
- D: They are mathematically related

EXAM TRAP: Lower WCSS → higher BCSS (better separation)!

Q26: Scree Plot Definition - Tutorial

QUESTION: What is a scree plot in K-means clustering?

OPTIONS:

1. Plot of WCSS vs number of clusters (K)
2. Plot of data points in 2D
3. Plot of centroids
4. Plot of silhouette scores vs K

ANSWER: A

DETAILED EXPLANATION:

- x-axis: number of clusters (K)
- y-axis: total WCSS (inertia)
- Used to find optimal K via "elbow method"
- Named after scree in PCA

WHY OTHER OPTIONS ARE WRONG:

- B: This is a scatter plot
- C: This is centroid visualization
- D: This is silhouette plot

EXAM TRAP: Scree plot = WCSS vs K!

Q27: Elbow Method - Tutorial

QUESTION: In the elbow method, what is the "elbow" point?

OPTIONS:

1. The point where WCSS is lowest
2. The point where adding clusters gives diminishing returns
3. The point where WCSS starts increasing
4. The point with highest WCSS

ANSWER: B

DETAILED EXPLANATION:

- Rate of WCSS decrease sharply changes
- Before elbow: rapid decrease
- After elbow: slow decrease
- Represents trade-off between fit and complexity

WHY OTHER OPTIONS ARE WRONG:

- A: WCSS is lowest at highest K
- C: WCSS never increases with K
- D: Highest WCSS is at K=1

EXAM TRAP: Elbow = diminishing returns from adding clusters!

Q28: Elbow Method Example - Tutorial

QUESTION: In the stock/bond example, what was the optimal K?

OPTIONS:

1. K=1
2. K=2
3. K=3
4. K=4

ANSWER: B

DETAILED EXPLANATION:

- Scree plot showed sharp change at K=2
- Rate of WCSS decrease changed dramatically
- Two distinct clusters identified
- Corresponds to market regimes (bull/bear)

WHY OTHER OPTIONS ARE WRONG:

- A: K=1 has highest WCSS
- C: K=3 showed diminishing returns
- D: K=4 had minimal improvement

EXAM TRAP: Elbow at K=2 in stock/bond example!

Q29: Silhouette Score Definition - Tutorial

QUESTION: What does the silhouette score measure?

OPTIONS:

1. Distance from points to centroids
2. How similar a point is to its own cluster compared to others
3. The number of clusters
4. The overall variance

ANSWER: B

DETAILED EXPLANATION:

- Compares intra-cluster cohesion vs inter-cluster separation
- For each point: $s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$
- a_i: mean distance to points in own cluster
- b_i: mean distance to points in closest other cluster

WHY OTHER OPTIONS ARE WRONG:

- A: This is WCSS/distance to centroid
- C: This is K (number of clusters)
- D: This is total variance

EXAM TRAP: Silhouette score = cohesion vs separation!

Q30: Silhouette Score Interpretation - Tutorial

QUESTION: What does a silhouette score close to 1 indicate?

OPTIONS:

1. Poor clustering
2. Excellent clustering (points close to own cluster, far from others)
3. Overlapping clusters
4. All points are outliers

ANSWER: B

DETAILED EXPLANATION:

- $s \to 1$: points very close to own cluster centroid
- $s \to 0$: clusters significantly overlapping
- $s < 0$: points may be mis-clustered
- Range: [-1, 1]

WHY OTHER OPTIONS ARE WRONG:

- A: Close to 1 means GOOD clustering
- C: Overlapping clusters give scores near 0
- D: Outliers would have low or negative scores

EXAM TRAP: $s \approx 1 =$ excellent clustering!

QUESTION: What is the range of silhouette scores?

OPTIONS:

0, 1

-1, 1

finity

finity

ANSWER: B

DETAILED EXPLANATION:

- Range: [-1, 1]
- $s = 1$: perfect clustering (points at centroids)
- $s = 0$: overlapping clusters
- $s < 0$: mis-clustered points (empirically unlikely)

WHY OTHER OPTIONS ARE WRONG:

- A: Range includes negative values
- C: Range is bounded
- D: Range includes negative values

EXAM TRAP: Silhouette score range = [-1, 1]!

Q32: Silhouette Score to Choose K - Tutorial

QUESTION: What K is optimal based on silhouette scores?

OPTIONS:

1. The K with the lowest average silhouette score
2. The K with the highest average silhouette score
3. Any K with score above 0
4. The K where scores start decreasing

ANSWER: B

DETAILED EXPLANATION:

- Compute average silhouette score for each K
- Choose K with highest average score
- Higher score = better-defined clusters
- Alternative to elbow method

WHY OTHER OPTIONS ARE WRONG:

- A: Lowest score indicates poor clustering
- C: Need to compare across K values
- D: Choose highest, not where it starts decreasing

EXAM TRAP: Optimal K = highest silhouette score!

Q33: K-means Advantages - Tutorial

QUESTION: What is an advantage of K-means clustering?

OPTIONS:

1. It always finds the optimal solution
2. It is intuitive, computationally efficient, and guaranteed to converge
3. It handles non-spherical clusters well
4. It automatically determines the number of clusters

ANSWER: B

DETAILED EXPLANATION:

- **Intuitive:** Easy to understand and communicate
- **Computationally efficient:** Scales well to large datasets
- **Guaranteed convergence:** Always reaches a solution
- **Visualizable:** Works well with 2-3 features

WHY OTHER OPTIONS ARE WRONG:

- A: May find local optimum, not global
- C: K-means struggles with non-spherical clusters
- D: K must be specified beforehand

EXAM TRAP: K-means = intuitive, efficient, guaranteed convergence!

Q34: K-means Disadvantages - Tutorial

QUESTION: What is a disadvantage of K-means?

OPTIONS:

1. It requires K to be specified in advance
2. It works well with non-spherical clusters
3. It is robust to outliers
4. It handles high-dimensional data well

ANSWER: A

DETAILED EXPLANATION:

- **K must be pre-specified:** Analyst must choose K
- **Sensitive to outliers:** Can distort centroids
- **Curse of dimensionality:** Poor with many features
- **Spherical clusters:** Assumes roughly spherical clusters

WHY OTHER OPTIONS ARE WRONG:

- B: K-means struggles with non-spherical clusters
- C: K-means is sensitive to outliers
- D: K-means suffers from curse of dimensionality

EXAM TRAP: K-means disadvantages = requires K, sensitive to outliers, spherical assumption!

Q35: K-means and Spherical Clusters - Tutorial

QUESTION: What cluster shape does K-means tend to produce?

OPTIONS:

1. Arbitrary shapes
2. Spherical (circular in 2D) clusters
3. Elongated clusters
4. Irregular clusters

ANSWER: B

DETAILED EXPLANATION:

- Based on Euclidean distances from centroids
- Creates roughly spherical clusters
- Assumes clusters are isotropic
- Fails when clusters have non-spherical shapes

WHY OTHER OPTIONS ARE WRONG:

- A: K-means cannot handle arbitrary shapes well
- C: K-means struggles with elongated clusters
- D: Irregular clusters cause problems

EXAM TRAP: K-means produces spherical clusters!

Q36: Outlier Sensitivity - Tutorial

QUESTION: How do outliers affect K-means?

OPTIONS:

1. They have no effect
2. They can distort centroids or form their own clusters
3. They improve clustering quality
4. They are automatically removed

ANSWER: B

DETAILED EXPLANATION:

- Outliers can pull centroids away from true centers
- In extreme cases, outliers may form their own clusters
- Particularly problematic with large K (overfitting)
- Scaling helps but may not eliminate issue

WHY OTHER OPTIONS ARE WRONG:

- A: Outliers have significant effect
- C: Outliers degrade clustering quality
- D: Outliers are not automatically removed

EXAM TRAP: Outliers distort centroids in K-means!

Q37: Curse of Dimensionality - Tutorial

QUESTION: What is the "curse of dimensionality" in K-means?

OPTIONS:

1. Too many data points
2. As features increase, data becomes sparse and distances become similar
3. Too few features
4. Too many clusters

ANSWER: B

DETAILED EXPLANATION:

- Data points become increasingly spread out
- Distances between points and centroids become similar
- Harder to identify distinct clusters
- K-means converges very slowly

WHY OTHER OPTIONS ARE WRONG:

- A: Curse is about features, not observations
- C: Curse is about too MANY features
- D: Curse is about features, not clusters

EXAM TRAP: Curse of dimensionality = many features make distances similar!

Q38: Curse of Dimensionality Solution - Tutorial

QUESTION: How can the curse of dimensionality be addressed?

OPTIONS:

1. Use more data points
2. Remove features or use PCA to reduce dimensions
3. Use more clusters
4. Use a smaller distance measure

ANSWER: B

DETAILED EXPLANATION:

- **Remove features:** Eliminate irrelevant features
- **PCA:** Transform to smaller subset of components
- **Alternative distances:** Use cosine distance
- **Feature selection:** Keep only important features

WHY OTHER OPTIONS ARE WRONG:

- A: More data doesn't solve high dimensionality
- C: More clusters worsens the problem
- D: Distance measure choice is a consideration

EXAM TRAP: Solutions = remove features or use PCA!

Q39: Non-Spherical Clusters Problem - Tutorial

QUESTION: What is a pathological case for K-means?

OPTIONS:

1. Data with two spherical clusters
2. Data with concentric rings or elongated clusters
3. Data with outliers removed
4. Data with equal cluster sizes

ANSWER: B

DETAILED EXPLANATION:

- K-means fails with non-spherical shapes
- Example: Two concentric rings (inner and outer)
- Example: Elongated clusters (oval shapes)
- K-means cannot form appropriate clusters

WHY OTHER OPTIONS ARE WRONG:

- A: K-means works well with spherical clusters
- C: Outliers should be handled separately
- D: Equal sizes are not a problem

EXAM TRAP: K-means fails with non-spherical shapes like concentric rings!

Q40: Concentric Rings Example - Tutorial

QUESTION: Why does K-means fail with concentric rings?

OPTIONS:

1. Centroids would be in the same place
2. There are too many points
3. Distances to centroids are equal
4. There is no data

ANSWER: A

DETAILED EXPLANATION:

- Optimal centroids would be at center for both clusters
- Both centroids would be in the same position
- Distances to each centroid would be identical
- Algorithm cannot form two distinct clusters

WHY OTHER OPTIONS ARE WRONG:

- B: Number of points doesn't cause failure
- C: Distances are similar, not equal
- D: There is data in both rings

EXAM TRAP: Concentric rings = centroids would be in same place!

Q41: Elongated Clusters Problem - Tutorial

QUESTION: How does K-means misallocate points in elongated clusters?

OPTIONS:

1. Points at the edge may be closer to the other cluster's centroid
2. All points are correctly allocated
3. Only outliers are misallocated
4. No misallocation occurs

ANSWER: A

DETAILED EXPLANATION:

- Elongated clusters have points far from their centroid
- Points at the edge may be closer to other cluster's centroid
- Results in misallocation
- Example: Orange point in oval cluster

WHY OTHER OPTIONS ARE WRONG:

- B: Misallocation does occur
- C: Not just outliers are misallocated
- D: Misallocation is a known problem

EXAM TRAP: Elongated clusters cause edge points to be misallocated!

Q42: K-means++ - Tutorial

QUESTION: What is K-means++?

OPTIONS:

1. A faster version of K-means
2. An initialization method that spreads centroids far apart
3. A version that uses Manhattan distance
4. A version that automatically determines K

ANSWER: B

DETAILED EXPLANATION:

- Smarter initialization than random
- First centroid chosen randomly from data points
- Subsequent centroids chosen far from existing ones
- Probability proportional to squared distance

WHY OTHER OPTIONS ARE WRONG:

- A: It's about initialization, not speed
- C: Uses Euclidean distance by default
- D: K still must be specified

EXAM TRAP: K-means++ = spreads initial centroids apart!

Q43: K-means++ Advantages - Tutorial

QUESTION: What is the benefit of using K-means++?

OPTIONS:

1. It always finds the global optimum
2. It improves clustering results and speeds convergence
3. It eliminates the need for scaling
4. It works with any number of clusters

ANSWER: B

DETAILED EXPLANATION:

- Better initial centroids = better final clusters
- Reduces chance of poor local optima
- Often converges more quickly
- Reduces sensitivity to initialization

WHY OTHER OPTIONS ARE WRONG:

- A: No method guarantees global optimum
- C: Scaling is still needed
- D: K must still be specified

EXAM TRAP: K-means++ = better initialization, faster convergence!

Q44: Multiple Initializations - Tutorial

QUESTION: Why is K-means often run multiple times?

OPTIONS:

1. To increase computation time
2. Because initial centroids are random, different runs give different results
3. To use more data
4. To automatically determine K

ANSWER: B

DETAILED EXPLANATION:

- Random initialization affects final solution
- Different runs may find different local optima
- Choose run with lowest inertia (WCSS)
- Standard practice to run multiple times

WHY OTHER OPTIONS ARE WRONG:

- A: Not the purpose of multiple runs
- C: Data is the same for each run
- D: K must still be specified

EXAM TRAP: Multiple runs needed because initialization affects results!

Q45: Choosing Between Initializations - Tutorial

QUESTION: How to choose between different K-means initializations?

OPTIONS:

1. Choose the one with highest WCSS
2. Choose the one with lowest inertia (WCSS)
3. Choose the one with most clusters
4. Choose the one with fastest convergence

ANSWER: B

DETAILED EXPLANATION:

- Lower WCSS = better fit
- More compact clusters
- Represents best clustering solution
- Standard practice in K-means

WHY OTHER OPTIONS ARE WRONG:

- A: Higher WCSS means poorer fit
- C: K is fixed across runs
- D: Convergence speed is secondary

EXAM TRAP: Choose initialization with lowest WCSS!

Q46: K-means Rule of Thumb - Tutorial

QUESTION: What is a rough rule of thumb for choosing K?

OPTIONS:

1. $K = \sqrt{N}$
2. $K \approx \sqrt{N/2}$
3. $K = N/2$
4. $K = \log(N)$

ANSWER: B

DETAILED EXPLANATION:

- Rule of thumb: $K \approx \sqrt{N/2}$
- Very rough guideline only
- Doesn't consider data characteristics
- Better to use elbow method or silhouette scores

WHY OTHER OPTIONS ARE WRONG:

- A: This is a different rule
- C: This is too large
- D: This is too small

EXAM TRAP: Rule of thumb: $K \approx \sqrt{N/2}$!

Q47: K=1 in K-means - Tutorial

QUESTION: What happens when K=1 in K-means?

OPTIONS:

1. Each point is its own cluster
2. All points in one cluster
3. The algorithm fails
4. Clusters are well-separated

ANSWER: B

DETAILED EXPLANATION:

- All data points in a single cluster
- Centroid is the global mean
- WCSS is total variance
- Usually not a meaningful solution

WHY OTHER OPTIONS ARE WRONG:

- A: This occurs when K=N
- C: K=1 is valid but uninformative
- D: With K=1, there is only one cluster

EXAM TRAP: K=1 = all points in one cluster!

Q48: K=N in K-means - Tutorial

QUESTION: What happens when K=N in K-means?

OPTIONS:

1. All points in one cluster
2. Each point is its own cluster
3. The algorithm fails
4. Clusters are optimal

ANSWER: B

DETAILED EXPLANATION:

- Each data point is its own cluster
- WCSS $= 0$ (perfect fit)
- Entirely uninformative
- Classic case of overfitting

WHY OTHER OPTIONS ARE WRONG:

- A: This happens when K=1
- C: Algorithm works but is useless
- D: K=N is overfitting, not optimal

EXAM TRAP: K=N = each point is its own cluster (overfitting)!

Q49: Overfitting in K-means - Tutorial

QUESTION: When does overfitting occur in K-means?

OPTIONS:

1. When K is too small
2. When K is too large (approaching N)
3. When features are scaled
4. When K=2

ANSWER: B

DETAILED EXPLANATION:

- Too many clusters = K close to N
- Each point becomes its own cluster
- WCSS approaches zero
- Model is uninformative

WHY OTHER OPTIONS ARE WRONG:

- A: K too small is underfitting
- C: Scaling prevents bias, not overfitting
- D: K=2 is usually not overfitting

EXAM TRAP: Overfitting = K too large (approaching N)!

Q50: K-means Performance Summary - Tutorial

QUESTION: What is the most comprehensive summary of K-means?

OPTIONS:

1. It always works perfectly
2. It is intuitive, efficient, but has limitations (requires K, sensitive to outliers, spherical assumption)
3. It is only useful for small datasets
4. It has no limitations

ANSWER: B

DETAILED EXPLANATION:

- **Advantages:** Intuitive, efficient, guaranteed convergence
- **Limitations:** Requires K, sensitive to outliers, spherical assumption
- **Applications:** Customer segmentation, anomaly detection
- **Evaluation:** WCSS, elbow method, silhouette scores

WHY OTHER OPTIONS ARE WRONG:

- A: K-means has limitations
- C: K-means scales well to large datasets
- D: K-means has several limitations

EXAM TRAP: K-means = powerful but with important limitations!

Q51: Hierarchical Clustering Advantage - Tutorial

QUESTION: What is a key advantage of hierarchical clustering over K-means?

OPTIONS:

1. It is faster than K-means
2. It does not require pre-specifying the number of clusters
3. It always produces spherical clusters
4. It handles outliers better

ANSWER: B

DETAILED EXPLANATION:

- No need to specify K in advance
- Builds hierarchy of all possible clusters
- From 1 cluster to N clusters
- Dendrogram shows all levels

WHY OTHER OPTIONS ARE WRONG:

- A: Hierarchical is often slower for large datasets
- C: Can handle various shapes
- D: DBSCAN handles outliers better

EXAM TRAP: Hierarchical clustering = no need to pre-specify K!

Q52: Dendrogram Definition - Tutorial

QUESTION: What is a dendrogram in hierarchical clustering?

OPTIONS:

1. A scatter plot of data points
2. A tree-like diagram showing cluster hierarchy
3. A bar chart of cluster sizes
4. A plot of WCSS vs K

ANSWER: B

DETAILED EXPLANATION:

- Visual representation of hierarchical clustering
- Tree-like structure
- Data points on x-axis
- Distance/dissimilarity on y-axis
- Shows merging or splitting process

WHY OTHER OPTIONS ARE WRONG:

- A: This is a scatter plot
- C: This is a bar chart
- D: This is a scree plot

EXAM TRAP: Dendrogram = tree diagram of cluster hierarchy!

Q53: Dendrogram Interpretation - Tutorial

QUESTION: What do the heights of vertical lines in a dendrogram represent?

OPTIONS:

1. Number of data points
2. Distance at which clusters are merged or split
3. Cluster sizes
4. Feature values

ANSWER: B

DETAILED EXPLANATION:

- Vertical lines show merge distance
- Longer lines = greater cluster dissimilarity
- Higher merges = less similar clusters
- Helps determine appropriate number of clusters

WHY OTHER OPTIONS ARE WRONG:

- A: Number of points shown on x-axis
- C: Size is shown by branch thickness
- D: Feature values are not shown

EXAM TRAP: Dendrogram height = distance at which clusters merge/split!

QUESTION: How does single linkage define distance between clusters?

OPTIONS:

1. Distance between farthest points in clusters
2. Distance between closest points in clusters
3. Distance between centroids
4. Average distance between all points

ANSWER: B

DETAILED EXPLANATION:

- $d_{single}(A, B) = \min_{a \in A, b \in B} d(a, b)$
- Based on closest points between clusters
- Also called nearest-neighbor linkage
- Tends to produce elongated clusters

WHY OTHER OPTIONS ARE WRONG:

- A: This is complete linkage
- C: This is centroid linkage
- D: This is average linkage

EXAM TRAP: Single linkage = closest points between clusters!

Q55: Complete Linkage - Tutorial

QUESTION: How does complete linkage define distance between clusters?

OPTIONS:

1. Distance between closest points in clusters
2. Distance between farthest points in clusters
3. Distance between centroids
4. Average distance between all points

ANSWER: B

DETAILED EXPLANATION:

- $d_{complete}(A, B) = \max_{a \in A, b \in B} d(a, b)$
- Based on farthest points between clusters
- Also called farthest-neighbor linkage
- Tends to produce compact, spherical clusters

WHY OTHER OPTIONS ARE WRONG:

- A: This is single linkage
- C: This is centroid linkage
- D: This is average linkage

EXAM TRAP: Complete linkage = farthest points between clusters!

Q56: Single vs Complete Linkage - Tutorial

QUESTION: How do single and complete linkage differ?

OPTIONS:

1. Single uses closest points; complete uses farthest points
2. Single uses farthest points; complete uses closest points
3. Both use the same method
4. Single is faster than complete

ANSWER: A

DETAILED EXPLANATION:

- **Single:** closest points (nearest neighbors)
- **Complete:** farthest points (most distant)
- **Single:** elongated clusters possible
- **Complete:** spherical clusters

WHY OTHER OPTIONS ARE WRONG:

- B: This is reversed
- C: They are different methods
- D: Speed difference is not the key distinction

EXAM TRAP: Single = closest; Complete = farthest!

Q57: Hierarchical Clustering Example - Tutorial

QUESTION: In the hierarchical clustering example, what was the first merge?

OPTIONS:

1. A and D
2. B and C
3. E and F
4. A and B

ANSWER: B

DETAILED EXPLANATION:

- Smallest distance in distance matrix was 3
- Between objects B and C
- B and C merged first
- Distance at merge = 3

WHY OTHER OPTIONS ARE WRONG:

- A: A and D merged later (distance 4)
- C: E and F had distance 18
- D: A and B had distance 5

EXAM TRAP: First merge = closest pair (B and C)!

Q58: Distance Matrix - Tutorial

QUESTION: What does a distance matrix show in hierarchical clustering?

OPTIONS:

1. Similarity between each pair of data points
2. Distance between each pair of data points
3. Cluster assignments
4. Centroids of clusters

ANSWER: B

DETAILED EXPLANATION:

- Matrix of pairwise distances
- Used to identify closest clusters
- Symmetric matrix
- Diagonal entries are zero

WHY OTHER OPTIONS ARE WRONG:

- A: Distance, not similarity (though related)
- C: Cluster assignments are from algorithm
- D: Centroids are from K-means

EXAM TRAP: Distance matrix shows pairwise distances!

Q59: Distance Matrix Update - Tutorial

QUESTION: How is the distance matrix updated after merging clusters?

OPTIONS:

1. It is not updated
2. Using the chosen linkage criterion (single, complete, etc.)
3. By averaging all distances
4. By taking the maximum distance

ANSWER: B

DETAILED EXPLANATION:

- New distances computed using linkage criterion
- Single linkage: min distance
- Complete linkage: max distance
- Process repeats iteratively

WHY OTHER OPTIONS ARE WRONG:

- A: Matrix must be updated
- C: Depends on linkage criterion
- D: Only for complete linkage

EXAM TRAP: Distance matrix updated using linkage criterion!

Q60: Hierarchical vs K-means - Tutorial

QUESTION: What is a key difference between hierarchical and K-means clustering?

OPTIONS:

1. Hierarchical requires specifying K; K-means does not
2. Hierarchical does not require specifying K; K-means does
3. Both require specifying K
4. Neither requires specifying K

ANSWER: B

DETAILED EXPLANATION:

- K-means: K must be pre-specified
- Hierarchical: No need to pre-specify K
- Hierarchical builds complete hierarchy
- Can choose any K from dendrogram

WHY OTHER OPTIONS ARE WRONG:

- A: This is reversed
- C: K-means requires K, hierarchical does not
- D: K-means requires K

EXAM TRAP: Hierarchical = no K needed; K-means = K required!

Q61: Hierarchical Advantages - Tutorial

QUESTION: What are advantages of hierarchical clustering?

OPTIONS:

1. Faster than K-means
2. No need to specify K and reveals nested relationships
3. Always finds global optimum
4. Works well with high-dimensional data

ANSWER: B

DETAILED EXPLANATION:

- **No K needed:** Does not require pre-specifying clusters
- **Nested relationships:** Reveals hierarchical structure
- **Dendrogram:** Easy to interpret
- **Flexible:** Can choose any K from dendrogram

WHY OTHER OPTIONS ARE WRONG:

- A: Hierarchical is often slower
- C: No method guarantees global optimum
- D: Hierarchical also suffers from curse of dimensionality

EXAM TRAP: Hierarchical advantages = no K needed + nested relationships!

Q62: Choosing K from Dendrogram - Tutorial

QUESTION: How can K be chosen from a dendrogram?

OPTIONS:

1. Count the number of branches at the top
2. Make a horizontal cut at a chosen distance
3. Count the number of leaves
4. Use the elbow method

ANSWER: B

DETAILED EXPLANATION:

- Draw horizontal line at chosen distance
- Number of vertical lines it crosses = K
- Lower cut = more clusters
- Higher cut = fewer clusters

WHY OTHER OPTIONS ARE WRONG:

- A: Top has one cluster
- C: Leaves = N (all points)
- D: Elbow method is for K-means

EXAM TRAP: Choose K by cutting dendrogram at chosen height!

Q63: Long Vertical Lines in Dendrogram - Tutorial

QUESTION: What does a long vertical line in a dendrogram indicate?

OPTIONS:

1. Clusters are very similar
2. Clusters are very different (large distance between merges)
3. Many data points in the cluster
4. Few data points in the cluster

ANSWER: B

DETAILED EXPLANATION:

- Long vertical line = large merge distance
- Clusters being merged are very different
- Suggests adding this cluster is important
- Helps identify optimal K

WHY OTHER OPTIONS ARE WRONG:

- A: Short lines indicate similarity
- C: Number of points shown by branch thickness
- D: Number of points shown by branch thickness

EXAM TRAP: Long vertical line = large cluster separation!

Q64: Hierarchical with Distance Matrix - Tutorial

QUESTION: In agglomerative clustering, when does the algorithm stop?

OPTIONS:

1. When K clusters are formed
2. When all points are in one cluster
3. When WCSS is minimized
4. When centroids stabilize

ANSWER: B

DETAILED EXPLANATION:

- Starts with N clusters (each point)
- Merges until one cluster remains
- Complete hierarchy formed
- Stopping at one cluster is natural end

WHY OTHER OPTIONS ARE WRONG:

- A: Can stop at any K, but natural end is one cluster
- C: WCSS is for K-means
- D: Centroids are for K-means

EXAM TRAP: Agglomerative stops when all points in one cluster!

Q65: Dendrogram Leaf Order - Tutorial

QUESTION: What do the leaves in a dendrogram represent?

OPTIONS:

1. Clusters at final level
2. Individual data points
3. Centroids
4. Distance values

ANSWER: B

DETAILED EXPLANATION:

- Leaves represent individual observations
- Each leaf = one data point
- Labels on x-axis are data point names
- Internal nodes represent merged clusters

WHY OTHER OPTIONS ARE WRONG:

- A: Clusters are internal nodes
- C: Centroids are from K-means
- D: Distances are on y-axis

EXAM TRAP: Dendrogram leaves = individual data points!

Q66: DBSCAN Definition - Tutorial

QUESTION: What does DBSCAN stand for?

OPTIONS:

1. Density-Based Spatial Clustering of Applications with Noise
2. Distance-Based Spatial Clustering of Applications with Noise
3. Data-Based Spatial Clustering of Applications with Noise
4. Density-Based Spectral Clustering of Applications with Noise

ANSWER: A

DETAILED EXPLANATION:

- DBSCAN = Density-Based Spatial Clustering of Applications with Noise
- Popular density-based clustering algorithm
- Groups dense regions of data points
- Separated by low-density regions

WHY OTHER OPTIONS ARE WRONG:

- B: DBSCAN is density-based, not distance-based
- C: "Spatial" not "Spectral"
- D: DBSCAN is density-based

EXAM TRAP: DBSCAN = Density-Based Spatial Clustering of Applications with Noise!

Q67: DBSCAN Core Point - Tutorial

QUESTION: What is a core point in DBSCAN?

OPTIONS:

1. A point with no neighbors
2. A point with at least a minimum number of points within a radius
3. A point on the edge of a cluster
4. A point that is isolated

ANSWER: B

DETAILED EXPLANATION:

- Core point has $\geq$ MinPts points in its neighborhood
- Within radius ε (epsilon)
- Interior of dense region
- Forms the foundation of clusters

WHY OTHER OPTIONS ARE WRONG:

- A: This describes a noise point
- C: This describes a border point
- D: This describes a noise point

EXAM TRAP: Core point = dense neighborhood with $\geq$ MinPts!

Q68: DBSCAN Border Point - Tutorial

QUESTION: What is a border point in DBSCAN?

OPTIONS:

1. A point with at least a minimum number of points within a radius
2. A point on the edge/fringe of a cluster
3. A point with no neighbors
4. A point at the center of a cluster

ANSWER: B

DETAILED EXPLANATION:

- Not a core point itself
- Falls within radius of a core point
- On the fringe of a dense region
- Still part of the cluster

WHY OTHER OPTIONS ARE WRONG:

- A: This describes a core point
- C: This describes a noise point
- D: This describes a core point

EXAM TRAP: Border point = edge of cluster, not core!

QUESTION: What is a noise point in DBSCAN?

OPTIONS:

1. A point with at least a minimum number of points within a radius
2. A point on the edge of a cluster
3. A point that is neither core nor border
4. A point at the center of a cluster

ANSWER: C

DETAILED EXPLANATION:

- Does not meet core point criteria
- Not within radius of any core point
- In low-density region
- Treated as outlier/noise

WHY OTHER OPTIONS ARE WRONG:

- A: This describes a core point
- B: This describes a border point
- D: This describes a core point

EXAM TRAP: Noise point = neither core nor border!

Q70: DBSCAN Parameters - Tutorial

QUESTION: What are the two key parameters for DBSCAN?

OPTIONS:

1. K and number of iterations
2. Epsilon (radius) and MinPts (minimum points)
3. Number of features and number of clusters
4. WCSS and BCSS

ANSWER: B

DETAILED EXPLANATION:

- **Epsilon (ϵ):** Radius of neighborhood
- **MinPts:** Minimum points to form dense region
- Both must be specified by analyst
- Choice affects clustering results

WHY OTHER OPTIONS ARE WRONG:

- A: K is for K-means
- C: Features are data, not parameters
- D: WCSS/BCSS are evaluation metrics

EXAM TRAP: DBSCAN parameters = epsilon and MinPts!

QUESTION: What is an advantage of DBSCAN over K-means?

OPTIONS:

1. It requires specifying K
2. It can find clusters of arbitrary shapes
3. It is faster for all datasets
4. It requires scaling

ANSWER: B

DETAILED EXPLANATION:

- Can find non-spherical clusters
- Elongated, irregular shapes
- Concentric rings (K-means fails)
- Based on density, not distance to centroid

WHY OTHER OPTIONS ARE WRONG:

- A: DBSCAN does NOT require K
- C: DBSCAN can be slower
- D: DBSCAN still benefits from scaling

EXAM TRAP: DBSCAN = arbitrary shape clusters!

Q72: DBSCAN Outlier Handling - Tutorial

QUESTION: How does DBSCAN handle outliers?

OPTIONS:

1. They are forced into clusters
2. They are classified as noise and excluded
3. They are removed from the dataset
4. They become their own clusters

ANSWER: B

DETAILED EXPLANATION:

- Points in low-density regions = noise
- Automatically excluded from clusters
- Does not distort clusters
- Better than K-means for outlier handling

WHY OTHER OPTIONS ARE WRONG:

- A: DBSCAN does not force outliers into clusters
- C: Outliers are identified, not removed
- D: Outliers may become noise, not their own clusters

EXAM TRAP: DBSCAN outliers = noise (excluded from clusters)!

Q73: DBSCAN Limitation - Tutorial

QUESTION: What is a limitation of DBSCAN?

OPTIONS:

1. It requires specifying K
2. It struggles with clusters of widely varying densities
3. It always finds spherical clusters
4. It cannot handle outliers

ANSWER: B

DETAILED EXPLANATION:

- Single ϵ and MinPts for all clusters
- Different densities require different parameters
- May miss low-density clusters
- May merge high-density clusters

WHY OTHER OPTIONS ARE WRONG:

- A: DBSCAN does NOT require K
- C: DBSCAN finds arbitrary shapes
- D: DBSCAN handles outliers well

EXAM TRAP: DBSCAN struggles with varying density clusters!

Q74: DBSCAN vs K-means - Tutorial

QUESTION: When should DBSCAN be preferred over K-means?

OPTIONS:

1. When clusters are spherical and well-separated
2. When clusters have non-spherical shapes and outliers are present
3. When K is known
4. When data is low-dimensional

ANSWER: B

DETAILED EXPLANATION:

- Non-spherical shapes (elongated, irregular)
- Outliers/noise in the data
- Unknown number of clusters
- Examples: geographical data, anomaly detection

WHY OTHER OPTIONS ARE WRONG:

- A: K-means works well for spherical clusters
- C: DBSCAN doesn't require K
- D: Both can work with low-dimensional data

EXAM TRAP: DBSCAN for non-spherical shapes + outliers!

Q75: Density-Based Clustering Summary - Tutorial

QUESTION: What is the key concept behind density-based clustering?

OPTIONS:

1. Distance to centroids
2. Dense regions separated by low-density regions
3. Hierarchical merging
4. Predefined number of groups

ANSWER: B

DETAILED EXPLANATION:

- Clusters = dense regions of data points
- Separated by regions of lower density
- Can find arbitrary shapes
- Robust to outliers

WHY OTHER OPTIONS ARE WRONG:

- A: This is K-means/centroid-based
- C: This is hierarchical
- D: This is partitional

EXAM TRAP: Density-based = dense regions separated by low density!

Q76: Clustering Evaluation Challenge - Tutorial

QUESTION: Why is evaluating clustering more challenging than supervised learning?

OPTIONS:

1. Clustering is computationally more expensive
2. There is no predefined "right answer" or labels to compare against
3. Clustering requires more data
4. Clustering has fewer metrics available

ANSWER: B

DETAILED EXPLANATION:

- No known outcomes/labels
- Cannot compare to ground truth
- Clusters are subjective
- Need internal validation metrics

WHY OTHER OPTIONS ARE WRONG:

- A: Computational cost is not the key issue
- C: Data requirements vary
- D: Many metrics exist (WCSS, silhouette)

EXAM TRAP: Clustering has no predefined "right answer" to compare against!

Q77: Internal vs External Validation - Tutorial

QUESTION: What type of validation is used in clustering (no labels)?

OPTIONS:

1. External validation (comparing to labels)
2. Internal validation (using metrics like WCSS, silhouette)
3. Cross-validation
4. Hold-out validation

ANSWER: B

DETAILED EXPLANATION:

- Internal validation = uses only the data
- No external labels needed
- Metrics: WCSS, silhouette score, BCSS
- Assesses compactness and separation

WHY OTHER OPTIONS ARE WRONG:

- A: External validation requires labels (supervised)
- C: Cross-validation is for supervised learning
- D: Hold-out is for supervised learning

EXAM TRAP: Clustering uses internal validation (no labels)!

Q78: WCSS as Evaluation Metric - Tutorial

QUESTION: What does a lower WCSS indicate?

OPTIONS:

1. Worse clustering
2. Better clustering (more compact)
3. More clusters
4. Fewer clusters

ANSWER: B

DETAILED EXPLANATION:

- Lower WCSS = points closer to centroids
- Clusters are more compact
- Better fit to the data
- Goal is to minimize WCSS

WHY OTHER OPTIONS ARE WRONG:

- A: Lower WCSS means BETTER clustering
- C: WCSS typically decreases with more clusters
- D: WCSS doesn't directly indicate K

EXAM TRAP: Lower WCSS = better/more compact clusters!

Q79: WCSS Limitation - Tutorial

QUESTION: What is a limitation of using WCSS alone?

OPTIONS:

1. It always decreases with K, leading to potential overfitting
2. It always increases with K
3. It is difficult to calculate
4. It cannot be used with large datasets

ANSWER: A

DETAILED EXPLANATION:

- WCSS never increases with K
- Adding clusters always improves WCSS
- K=N gives WCSS=0 (overfitting)
- Must balance with interpretability

WHY OTHER OPTIONS ARE WRONG:

- B: WCSS decreases, not increases
- C: WCSS is straightforward to calculate
- D: WCSS works with large datasets

EXAM TRAP: WCSS always decreases with K → overfitting risk!

QUESTION: What is an advantage of silhouette scores over WCSS?

OPTIONS:

1. It always increases with K
2. It balances within-cluster and between-cluster distances
3. It is easier to calculate
4. It does not require scaling

ANSWER: B

DETAILED EXPLANATION:

- Considers both cohesion (within) and separation (between)
- Higher score = better clustering
- Does not monotonically increase with K
- Better at identifying optimal K

WHY OTHER OPTIONS ARE WRONG:

- A: Silhouette score may decrease
- C: WCSS is simpler to calculate
- D: Both require scaling

EXAM TRAP: Silhouette score balances cohesion and separation!

Q81: Silhouette Score Interpretation Values - Tutorial

QUESTION: What does a silhouette score of 0 indicate?

OPTIONS:

1. Perfect clustering
2. Clusters are significantly overlapping
3. Mis-clustered points
4. All points are outliers

ANSWER: B

DETAILED EXPLANATION:

- $s = 0$: clusters are overlapping
- Points are equally close to own and other clusters
- No clear separation
- Not good clustering

WHY OTHER OPTIONS ARE WRONG:

- A: $s = 1$ indicates perfect clustering
- C: $s < 0$ indicates mis-clustering
- D: Outliers would have low/negative scores

EXAM TRAP: $s = 0 =$ overlapping clusters!

QUESTION: How do elbow method and silhouette scores compare for choosing K?

OPTIONS:

1. They always agree
2. Elbow uses WCSS (inertia); silhouette uses cohesion vs separation
3. Elbow is always better
4. Silhouette is always better

ANSWER: B

DETAILED EXPLANATION:

- **Elbow:** Visual inspection of WCSS vs K
- **Silhouette:** Numerical score [-1, 1]
- Both can be used together
- May not always agree

WHY OTHER OPTIONS ARE WRONG:

- A: They may not agree
- C: Neither is universally better
- D: Neither is universally better

EXAM TRAP: Elbow = WCSS; Silhouette = cohesion vs separation!

Q83: Silhouette Plots - Tutorial

QUESTION: In a silhouette plot, what does a negative value indicate?

OPTIONS:

1. Well-clustered points
2. Points are likely mis-clustered
3. Perfect clustering
4. Clusters are well-separated

ANSWER: B

DETAILED EXPLANATION:

- Negative silhouette score ($s < 0$)
- Point is closer to other clusters than own
- Suggests mis-clustering
- Empirically unlikely but possible

WHY OTHER OPTIONS ARE WRONG:

- A: Negative means POOR clustering
- C: $s = 1$ is perfect
- D: Positive values indicate separation

EXAM TRAP: Negative silhouette = mis-clustered points!

Q84: Stock/Bond Clustering Example - Tutorial

QUESTION: In the stock/bond example, what did the data represent?

OPTIONS:

1. Daily stock prices
2. Annual value-weighted stock returns and Treasury bill yields (1927-2021)
3. Monthly bond yields
4. Quarterly earnings

ANSWER: B

DETAILED EXPLANATION:

- Annual data from 1927 to 2021
- Two features: stock returns and T-bill yields
- Two distinct clusters identified
- Corresponds to different market regimes

WHY OTHER OPTIONS ARE WRONG:

- A: Not daily prices
- C: Annual, not monthly
- D: Not quarterly earnings

EXAM TRAP: Stock/bond data = annual returns and yields 1927-2021!

Q85: Clustering Evaluation Summary - Tutorial

QUESTION: What is the most comprehensive approach to evaluate clustering?

OPTIONS:

1. Use only WCSS
2. Use only silhouette scores
3. Use multiple metrics (WCSS, silhouette, BCSS) and visualization
4. Use supervised learning metrics

ANSWER: C

DETAILED EXPLANATION:

- No single metric tells the whole story
- Combine WCSS (compactness), silhouette (cohesion/separation), BCSS
- Visualize clusters when possible
- Consider domain knowledge

WHY OTHER OPTIONS ARE WRONG:

- A: WCSS alone can lead to overfitting
- B: Silhouette alone may miss some aspects
- D: Supervised metrics require labels

EXAM TRAP: Use multiple metrics + visualization for comprehensive evaluation!

Q86: Choosing K in Unsupervised Learning - Tutorial

QUESTION: How do you choose the number of clusters in unsupervised learning?

OPTIONS:

1. Use a fixed K for all datasets
2. Use scree plot (elbow method) and/or silhouette scores
3. Always use K=2
4. Use the square root of N

ANSWER: B

DETAILED EXPLANATION:

- **Scree plot:** Look for elbow in WCSS vs K
- **Silhouette scores:** Choose K with highest score
- Use domain knowledge
- Combine multiple methods

WHY OTHER OPTIONS ARE WRONG:

- A: Optimal K varies by dataset
- C: K=2 may not be optimal
- D: Rule of thumb is only a rough guide

EXAM TRAP: Choose K using elbow method and/or silhouette scores!

Q87: K-means Steps - Tutorial

QUESTION: What are the correct steps for K-means clustering?

OPTIONS:

1. Scale features, choose K, initialize centroids, assign points, update centroids, repeat
2. Choose K, assign points, scale features, update centroids
3. Assign points, update centroids, choose K
4. Scale features, assign points, choose K, update centroids

ANSWER: A

DETAILED EXPLANATION:

1. Scale features (essential for distance-based)
2. Choose K (number of clusters)
3. Initialize K centroids
4. Assign points to nearest centroid
5. Update centroids (mean of assigned points)
6. Repeat until convergence

WHY OTHER OPTIONS ARE WRONG:

- B: Scaling should come BEFORE choosing K
- C: K must be chosen before assignment
- D: Scaling should come first

EXAM TRAP: Correct order: scale, choose K, initialize, assign, update, repeat!

Q88: Choosing Between Initializations - Tutorial

QUESTION: How to choose between multiple K-means initializations?

OPTIONS:

1. Choose the one with highest WCSS
2. Choose the one with lowest WCSS (inertia)
3. Choose the one with highest silhouette score
4. Choose the one with fastest convergence

ANSWER: B

DETAILED EXPLANATION:

- Lower WCSS = better fit
- More compact clusters
- Run multiple times with different initializations
- Select best based on lowest WCSS

WHY OTHER OPTIONS ARE WRONG:

- A: Higher WCSS = worse fit
- C: Silhouette can also be used, but WCSS is standard
- D: Quality matters more than speed

EXAM TRAP: Select initialization with lowest WCSS!

Q89: Scree Plot Interpretation - Tutorial

QUESTION: How do you use a scree plot to choose K?

OPTIONS:

1. Choose the K with the lowest WCSS
2. Look for the "elbow" where WCSS decrease slows
3. Choose the K with the highest WCSS
4. Choose K=2 always

ANSWER: B

DETAILED EXPLANATION:

- x-axis: K (number of clusters)
- y-axis: WCSS (inertia)
- Elbow: point where slope changes from steep to flat
- Diminishing returns after elbow

WHY OTHER OPTIONS ARE WRONG:

- A: WCSS always decreases with K (overfitting)
- C: Highest WCSS is at K=1
- D: Optimal K varies by dataset

EXAM TRAP: Elbow = point where WCSS decrease slows!

Q90: Hierarchical Types - Tutorial

QUESTION: What are the two types of hierarchical clustering?

OPTIONS:

1. Partitional and Density-based
2. Divisive (top-down) and Agglomerative (bottom-up)
3. Single and Complete linkage
4. K-means and DBSCAN

ANSWER: B

DETAILED EXPLANATION:

- **Divisive:** Top-down (split from one cluster)
- **Agglomerative:** Bottom-up (merge from individual points)
- Both create hierarchical structure
- Visualized using dendrogram

WHY OTHER OPTIONS ARE WRONG:

- A: These are different clustering categories
- C: These are linkage criteria
- D: These are different algorithms

EXAM TRAP: Hierarchical = divisive and agglomerative!

Q91: Hierarchical Advantages Detailed - Tutorial

QUESTION: What are advantages of hierarchical clustering?

OPTIONS:

1. No need to specify K, reveals nested relationships, dendrograms are interpretable
2. Always finds global optimum
3. Faster than K-means
4. Works well with high-dimensional data

ANSWER: A

DETAILED EXPLANATION:

- **No K needed:** Does not require pre-specifying clusters
- **Nested relationships:** Shows hierarchy of clusters
- **Dendrogram:** Easy to interpret visually
- **Flexible:** Can choose any K from dendrogram

WHY OTHER OPTIONS ARE WRONG:

- B: No method guarantees global optimum
- C: Hierarchical is often slower
- D: Hierarchical suffers from curse of dimensionality

EXAM TRAP: Hierarchical = no K, nested relationships, interpretable!

Q92: K-means Spherical Assumption - Tutorial

QUESTION: Is the statement "K-means with Euclidean distance can only be used when clusters are approximately spherical" true?

OPTIONS:

1. True
2. False
3. Only true with Manhattan distance
4. Only false with scaling

ANSWER: A

DETAILED EXPLANATION:

- K-means based on Euclidean distance
- Tends to produce spherical clusters
- Fails with elongated or irregular shapes
- Manhattan distance produces rhombus shapes

WHY OTHER OPTIONS ARE WRONG:

- B: Statement is true
- C: Manhattan also has shape limitations
- D: Scaling doesn't change spherical assumption

EXAM TRAP: K-means assumes approximately spherical clusters!

QUESTION: Is the statement "WCSS will never rise when K increases" true?

OPTIONS:

1. True
2. False
3. It rises when K is too large
4. It only rises when K=1

ANSWER: A

DETAILED EXPLANATION:

- WCSS cannot increase with K
- More clusters = better fit
- K=N gives WCSS=0
- Similar to R^2 in regression

WHY OTHER OPTIONS ARE WRONG:

- B: Statement is true
- C: WCSS always decreases or stays same
- D: Always true for all K

EXAM TRAP: WCSS never increases with K!

Q94: Dendrogram Interpretation Detailed - Tutorial

QUESTION: What is a dendrogram and how is it interpreted?

OPTIONS:

1. A bar chart of cluster sizes
2. A tree diagram showing cluster hierarchy; heights show merge distances
3. A scatter plot of data points
4. A plot of WCSS vs K

ANSWER: B

DETAILED EXPLANATION:

- Visual representation of hierarchical clustering
- Data points on x-axis
- Distance/dissimilarity on y-axis
- Heights show merge distances

WHY OTHER OPTIONS ARE WRONG:

- A: This is a bar chart
- C: This is a scatter plot
- D: This is a scree plot

EXAM TRAP: Dendrogram = tree diagram showing cluster hierarchy!

Q95: Centroid Calculation - Tutorial

QUESTION: Given three banks with features, how is the centroid calculated?

OPTIONS:

1. By taking the median of each feature
2. By taking the average (mean) of each feature across banks
3. By taking the maximum of each feature
4. By taking the minimum of each feature

ANSWER: B

DETAILED EXPLANATION:

- Centroid = mean (average) of all points in cluster
- Average each feature separately
- Example: $(1.2+6.0+0.5)/3 = 2.567$
- Similarly for each feature

WHY OTHER OPTIONS ARE WRONG:

- A: K-means uses mean, not median
- C: Max would be extreme
- D: Min would be extreme

EXAM TRAP: Centroid = mean (average) of feature values!

Q96: Bank Centroid Example - Tutorial

QUESTION: For three banks, what is the centroid coordinate?

OPTIONS:

1. (2.567, 12.333, 176.667)
2. (1.200, 5.000, 80.000)
3. (6.000, 25.000, 400.000)
4. (0.500, 7.000, 50.000)

ANSWER: A

DETAILED EXPLANATION:

- Bank A: (1.2, 5, 80)
- Bank B: (6.0, 25, 400)
- Bank C: (0.5, 7, 50)
- Average each coordinate: $((1.2+6.0+0.5)/3, (5+25+7)/3, (80+400+50)/3) = (2.567, 12.333, 176.667)$

WHY OTHER OPTIONS ARE WRONG:

- B: This is Bank A's values
- C: This is Bank B's values
- D: This is Bank C's values

EXAM TRAP: Centroid = average of all points in cluster!

QUESTION: Why use standardization vs normalization for clustering?

OPTIONS:

1. They are the same
2. Standardization gives mean 0, variance 1; normalization scales to [0,1]
3. Normalization gives mean 0; standardization scales to [0,1]
4. Neither is needed

ANSWER: B

DETAILED EXPLANATION:

- **Standardization:** mean=0, variance=1
- **Normalization:** scale to [0,1] range
- Both make features comparable
- Choice depends on data characteristics

WHY OTHER OPTIONS ARE WRONG:

- A: They are different methods
- C: This is reversed
- D: Scaling is essential for clustering

EXAM TRAP: Standardization = mean 0, variance 1; Normalization = [0,1] range!

QUESTION: How does clustering differ from classification?

OPTIONS:

1. Clustering is supervised; classification is unsupervised
2. Clustering is unsupervised; classification is supervised
3. Both are supervised
4. Both are unsupervised

ANSWER: B

DETAILED EXPLANATION:

- **Clustering:** Unsupervised (no labels)
- **Classification:** Supervised (known labels)
- Clustering discovers groups
- Classification assigns to known groups

WHY OTHER OPTIONS ARE WRONG:

- A: This is reversed
- C: Clustering is unsupervised
- D: Classification is supervised

EXAM TRAP: Clustering = unsupervised; Classification = supervised!

Q99: Clustering as Exploratory Analysis - Tutorial

QUESTION: Why use clustering before supervised learning?

OPTIONS:

1. To remove all features
2. To understand data structure and potentially improve feature engineering
3. To label the data
4. To increase data size

ANSWER: B

DETAILED EXPLANATION:

- Exploratory data analysis
- Understand inherent structure
- May reveal patterns
- Can improve feature engineering

WHY OTHER OPTIONS ARE WRONG:

- A: Clustering doesn't remove features
- C: Clustering doesn't label data
- D: Clustering doesn't increase data size

EXAM TRAP: Clustering helps understand structure before supervised learning!

Q100: Final Summary - Tutorial

QUESTION: What is the most comprehensive summary of unsupervised learning?

OPTIONS:

1. Only K-means is used for clustering
2. Unsupervised learning includes K-means, hierarchical, and DBSCAN, each with different strengths and evaluation methods
3. Only hierarchical clustering is used
4. Clustering doesn't require evaluation

ANSWER: B

DETAILED EXPLANATION:

- **K-means:** Partitional, centroid-based, requires K
- **Hierarchical:** Tree-based, no K needed, dendrogram
- **DBSCAN:** Density-based, arbitrary shapes, noise handling
- **Evaluation:** WCSS, elbow method, silhouette scores

WHY OTHER OPTIONS ARE WRONG:

- A: Multiple methods exist beyond K-means
- C: Multiple methods exist beyond hierarchical
- D: Evaluation is essential for clustering

EXAM SUCCESS: Understand K-means, hierarchical, DBSCAN, and evaluation methods!